



Computer Vision for Autonomous Vehicles

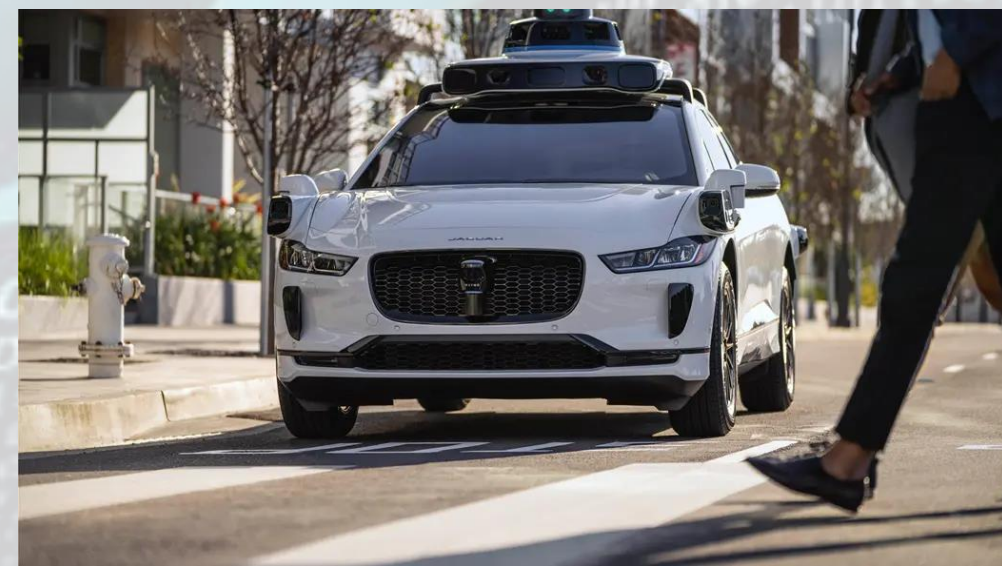
Quadrotor Control

Instructor: Golnaz Habibi, PhD

University of Oklahoma

Fall 2025

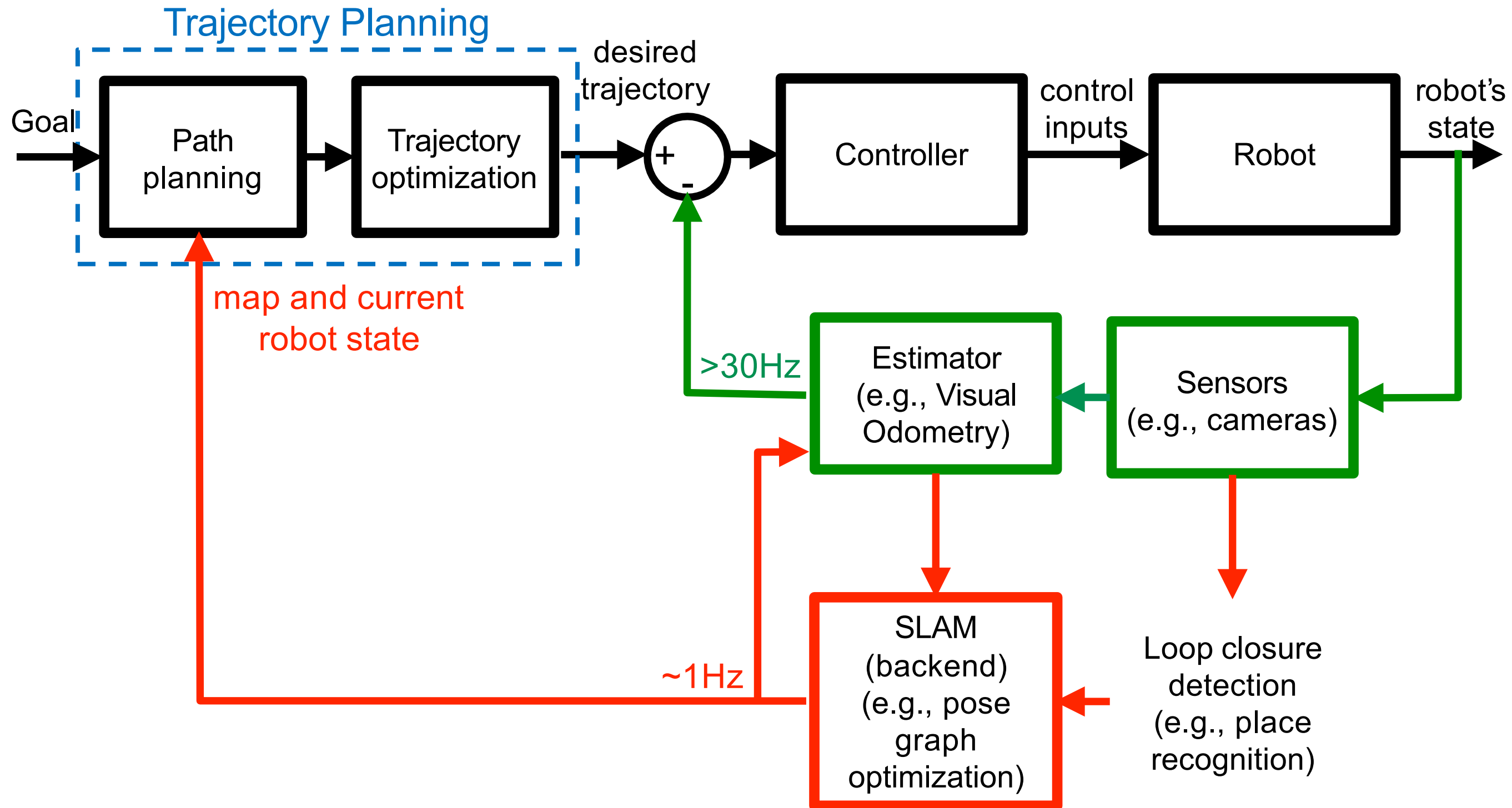
November 2025



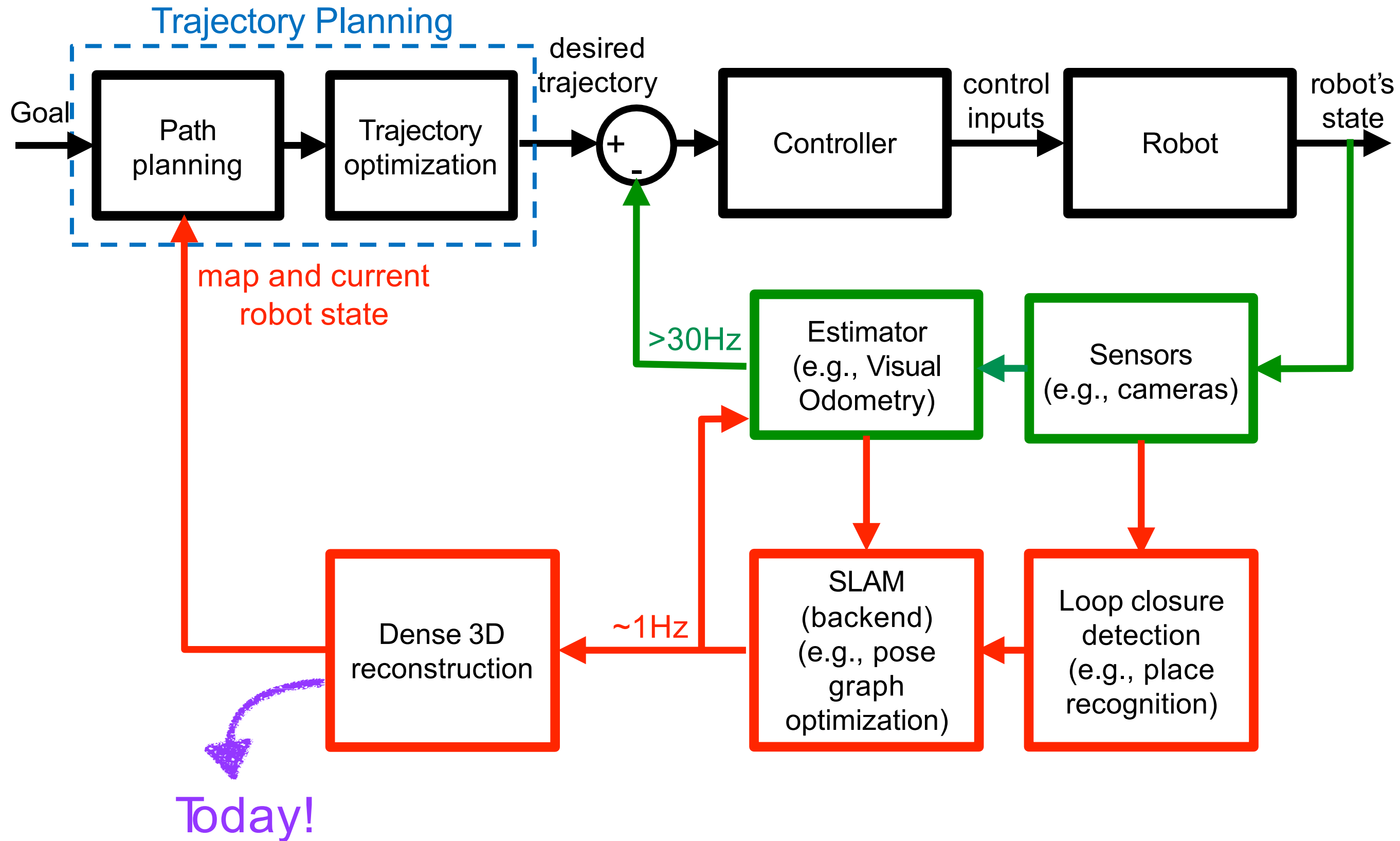
Announcement

- Lab 6 grade will be posted today
- Last Quiz solution will be posted today
- Final project presentation next Tuesday on December 16th, ALL member need to present
- Class survey by Sunday
- Today: summary of the class and 3D construction and RL

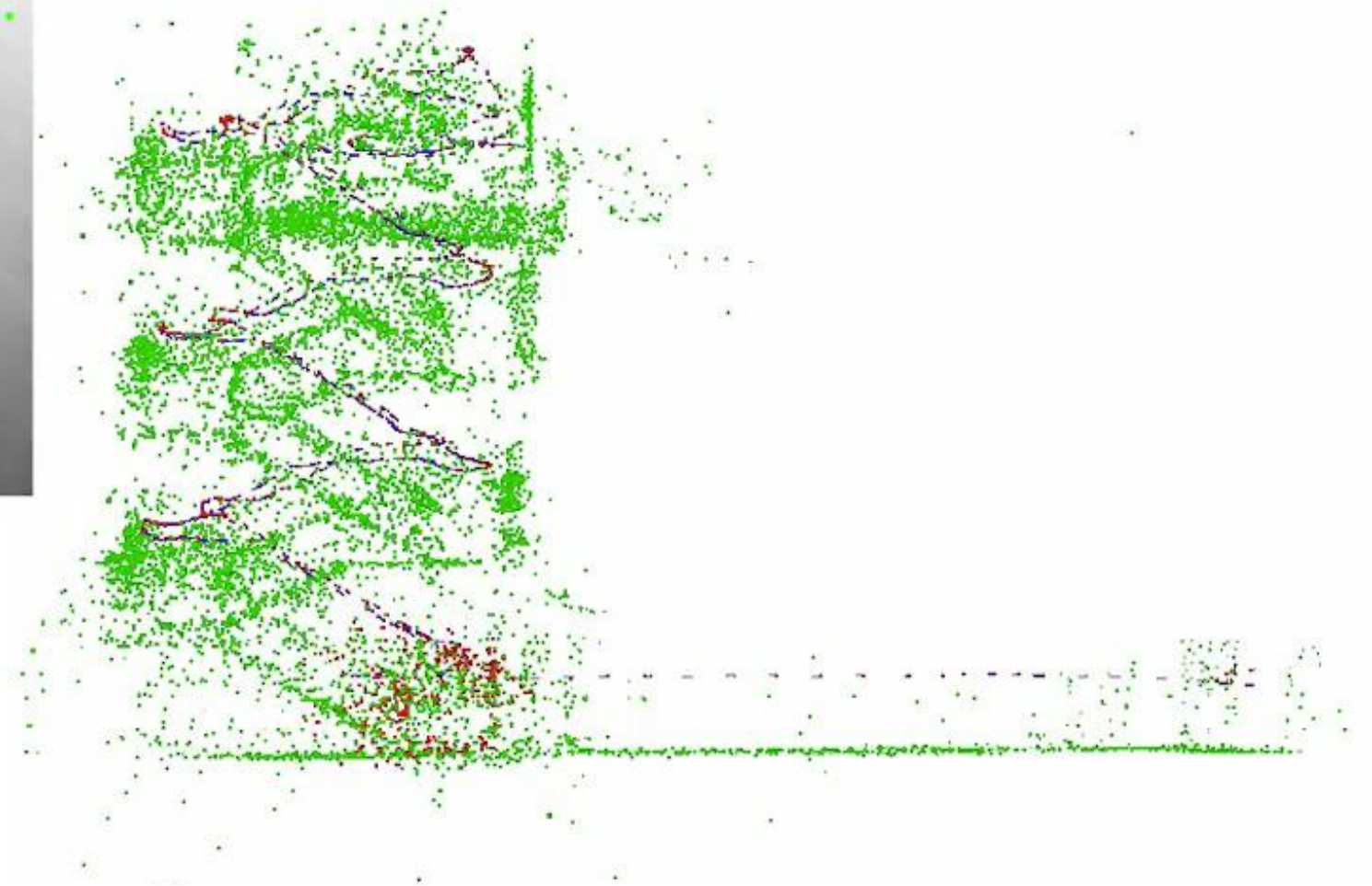
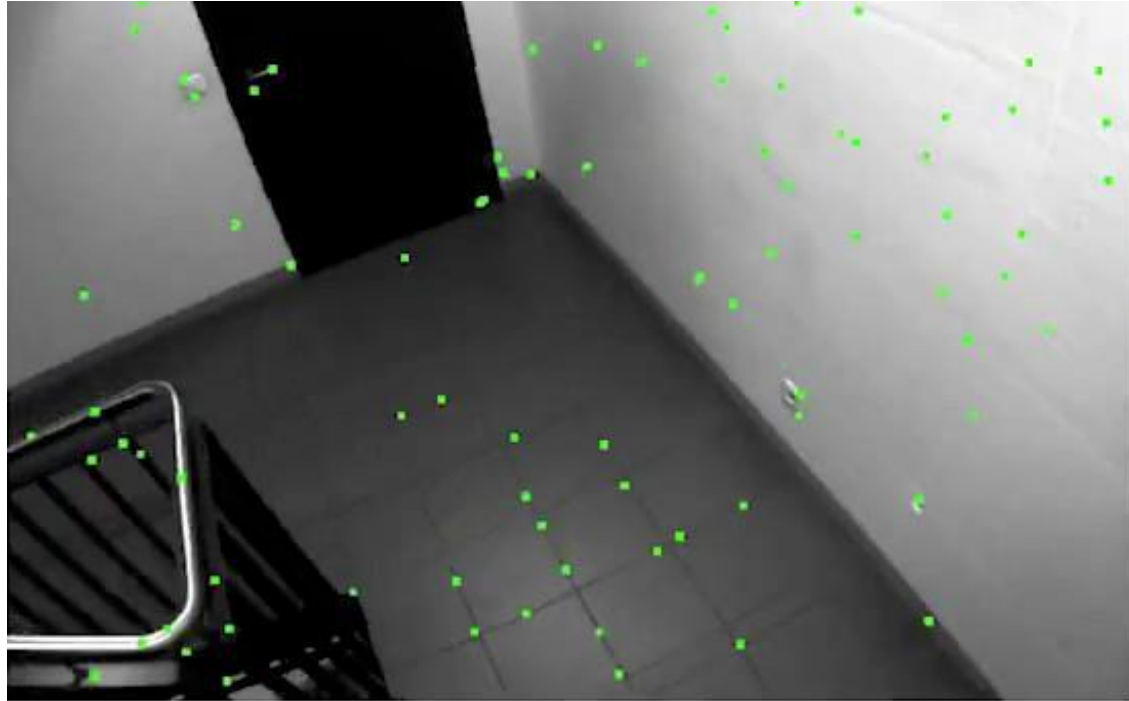
Big Picture



Big Picture



Point Clouds



Map
representation

3D Topology?

Lightweight?

Filters
Noise/Outliers?

Semantics?

Point Clouds

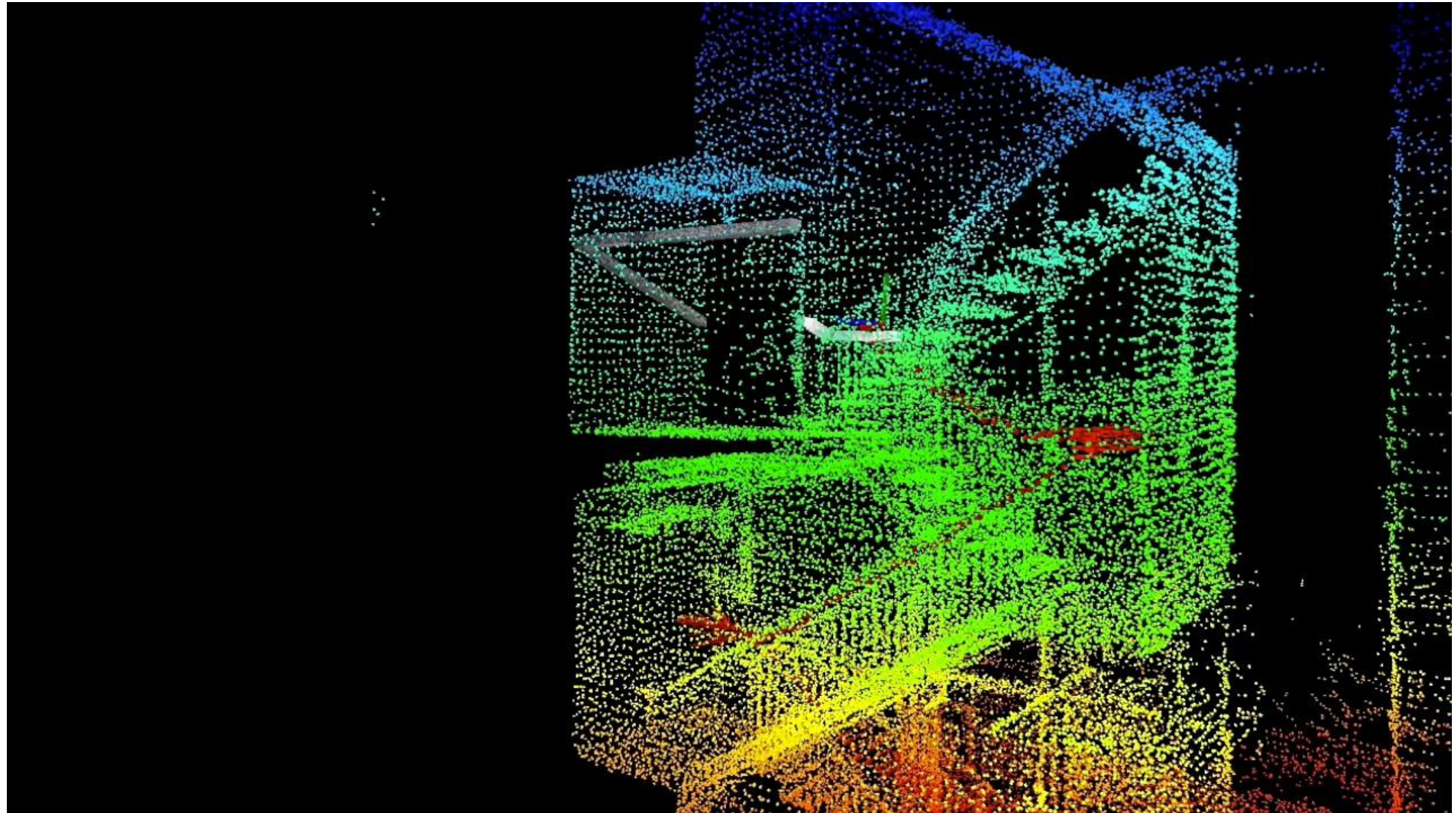
✗

✓/✗
No, if Dense

✗

✓/✗
No, if Sparse

Point Clouds

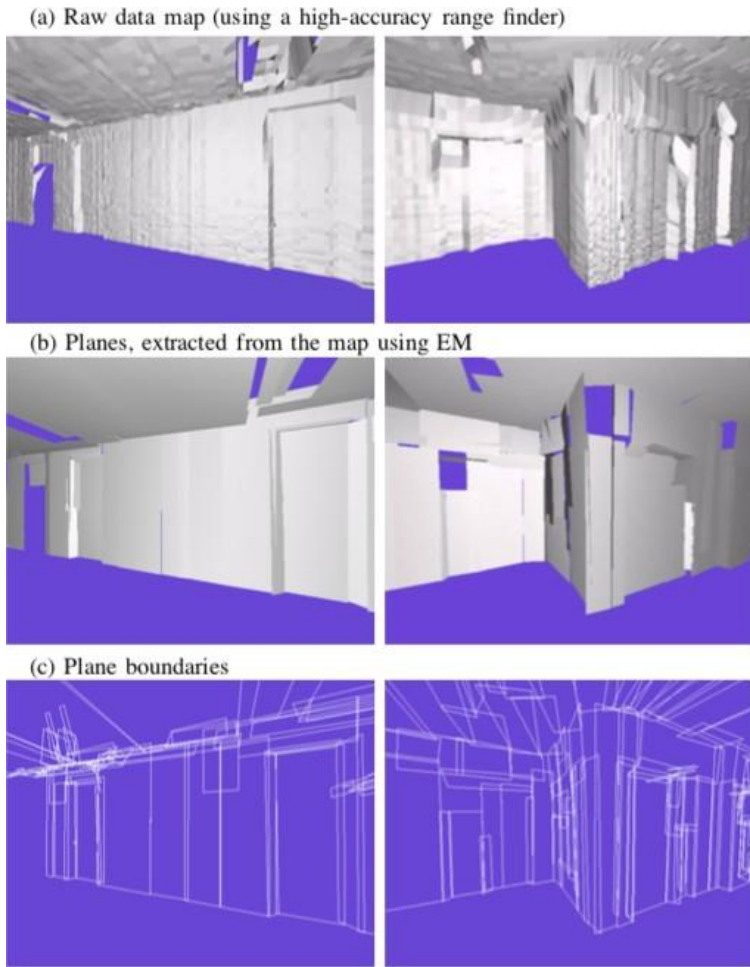


Map representation	3D Topology?	Lightweight?	Filters Noise/ Outliers?	Semantics?	Generality
Point Clouds	✗	✓/✗ No, if Dense	✗	✓/✗ No, if Sparse	✓

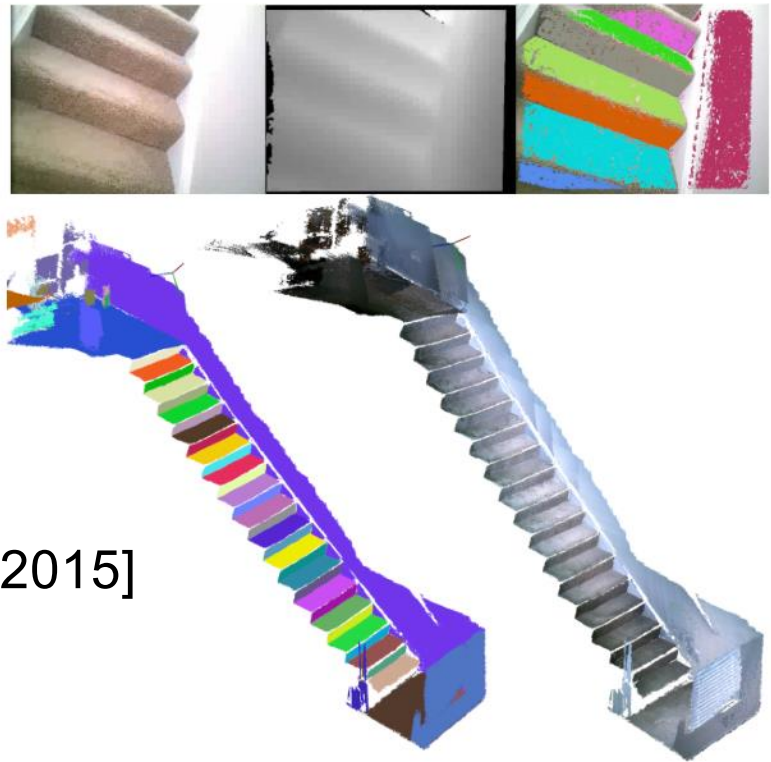
Geometric Primitives

Point, lines, planes

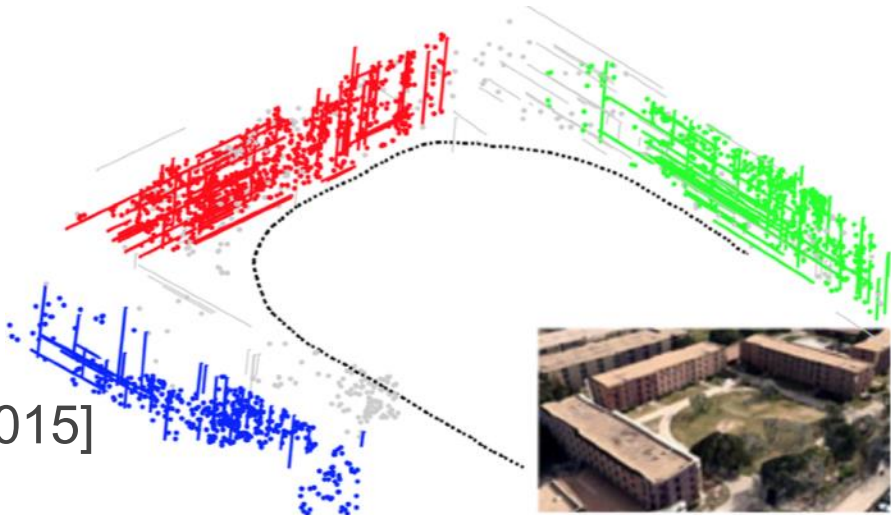
[Thrun et al. 2004]



[Kaess 2015]



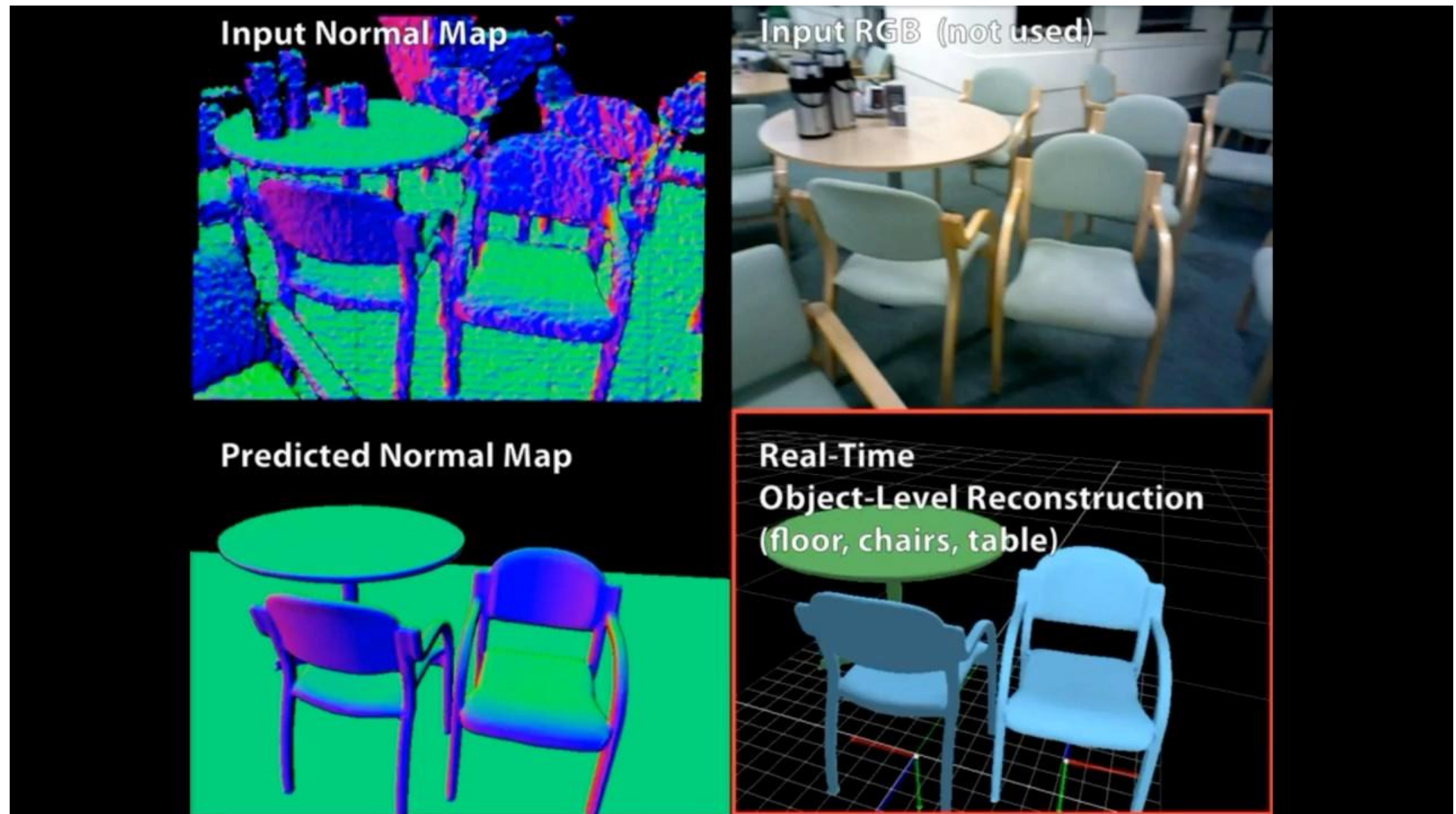
[Lu et al. 2015]



Map representation	3D Topology?	Lightweight?	Filters Noise/ Outliers?	Semantics?	Generality
Point Clouds	X	✓ No, if Dense	X	✓ No, if Sparse	✓
Geometric primitives	X	✓	✓	✓ No, if Sparse	X

Object-based Maps

[Salas-Moreno et al, 2014]

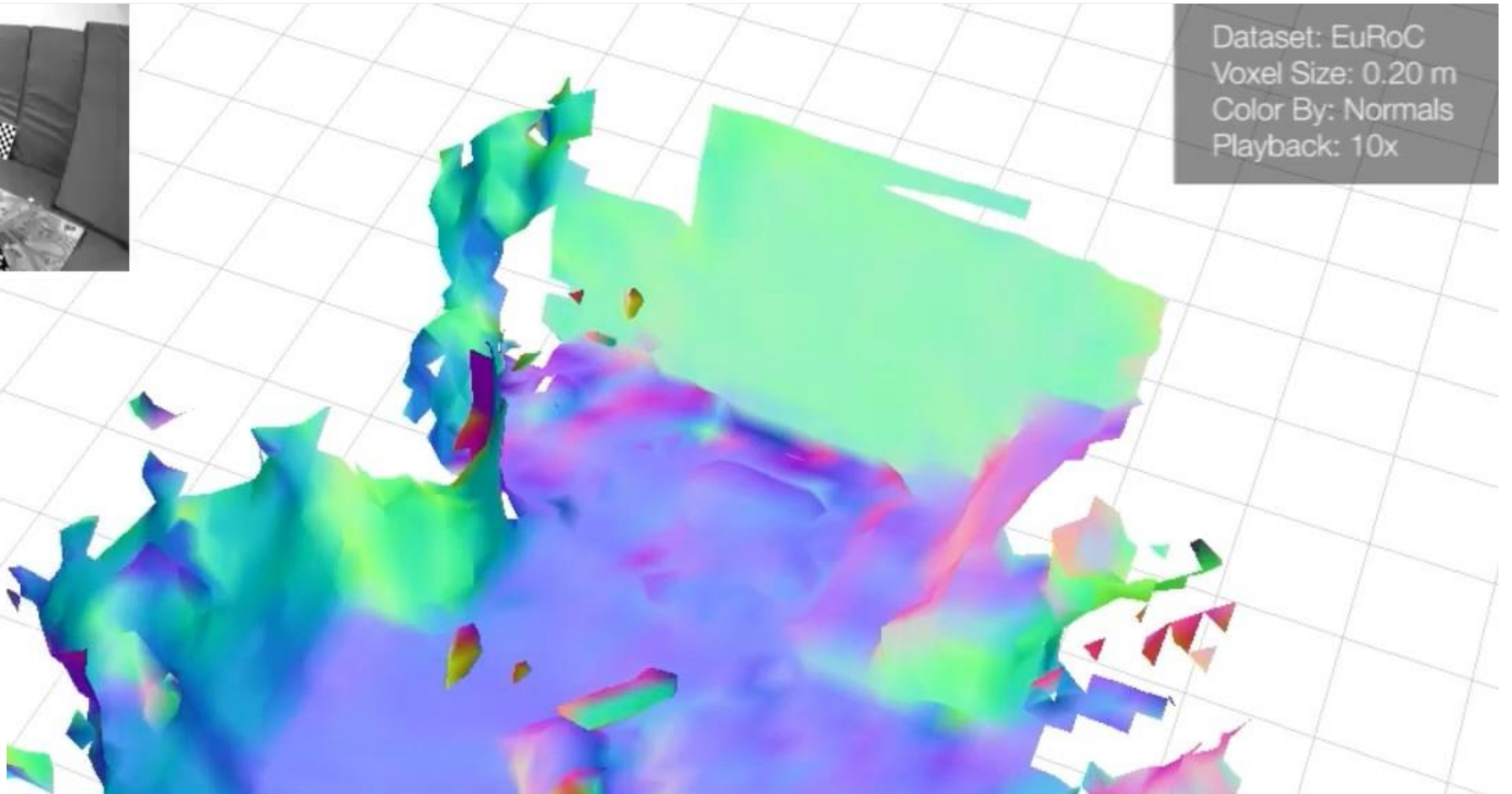


Map representation	3D Topology?	Lightweight?	Filters Noise/ Outliers?	Semantics?
Point Clouds	✗	✓/✗ No, if Dense	✗	✓/✗ No, if Sparse
primitives & objects	✗	✓	✓	✓/✗ No, if Sparse

Volumetric Methods: Voxels/Octrees



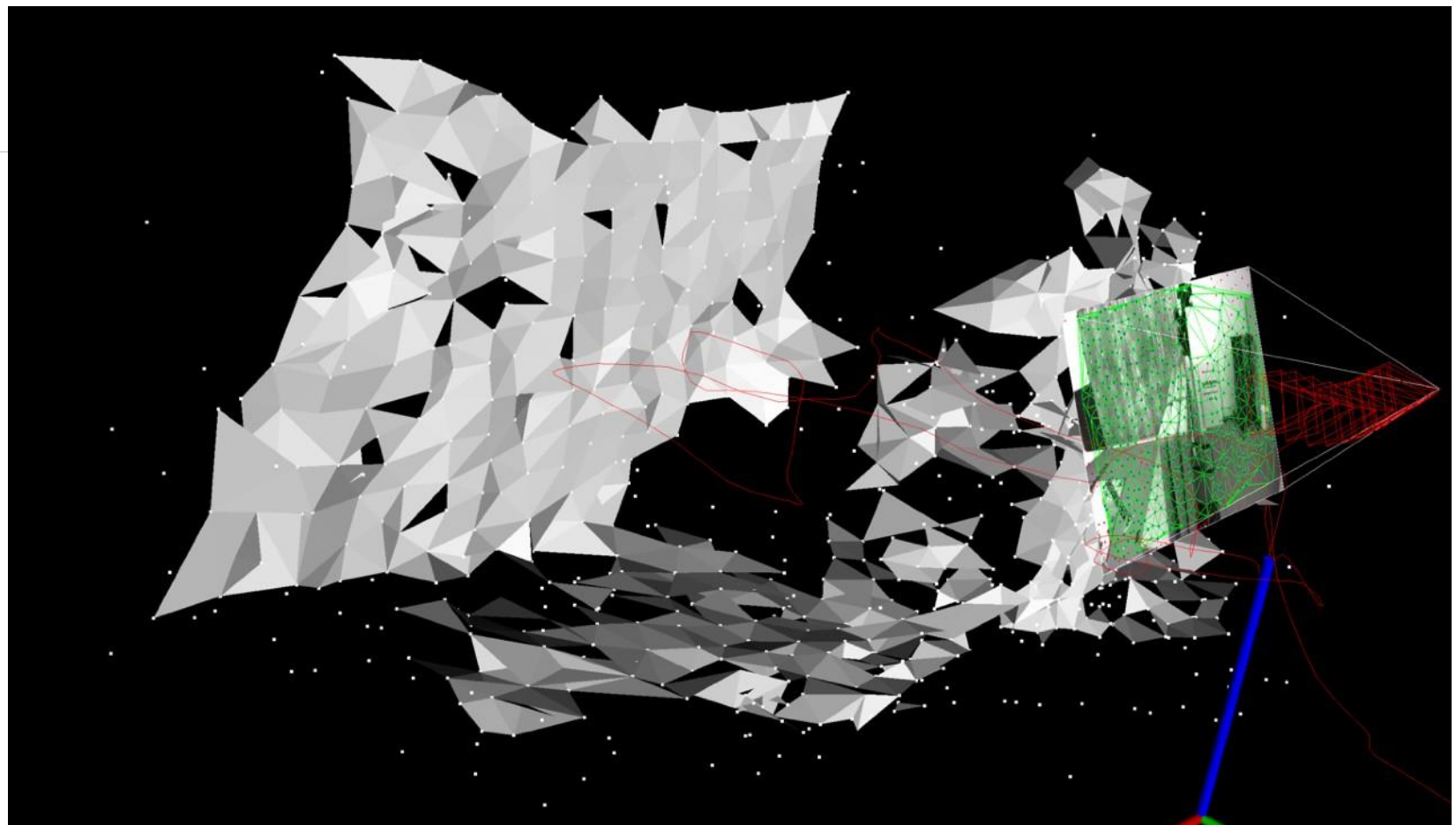
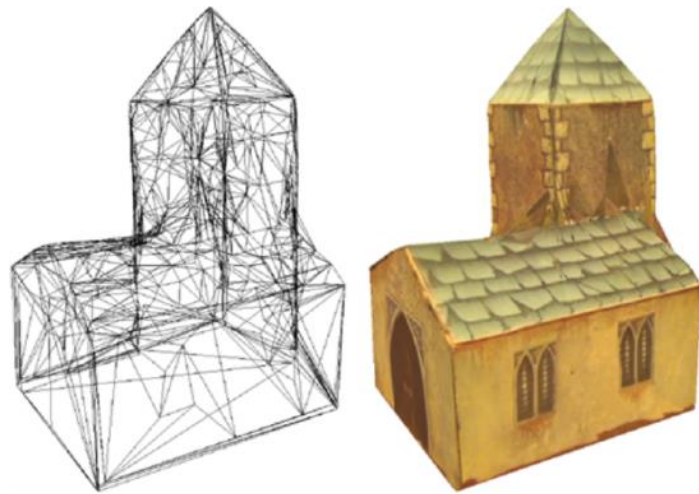
Dataset: EuRoC
Voxel Size: 0.20 m
Color By: Normals
Playback: 10x



[Oleynikova, ICRA'17]

Map representation	3D Topology?	Lightweight?	Filters Noise/Outliers?	Semantics?
Point Clouds	✗	✓/✗ No, if Dense	✗	✓/✗ No, if Sparse
primitives & objects	✗	✓	✓	✓/✗ No, if Sparse
Voxels	✓	✓/✗ No, if small voxel	✓	✓/✗ No, if large

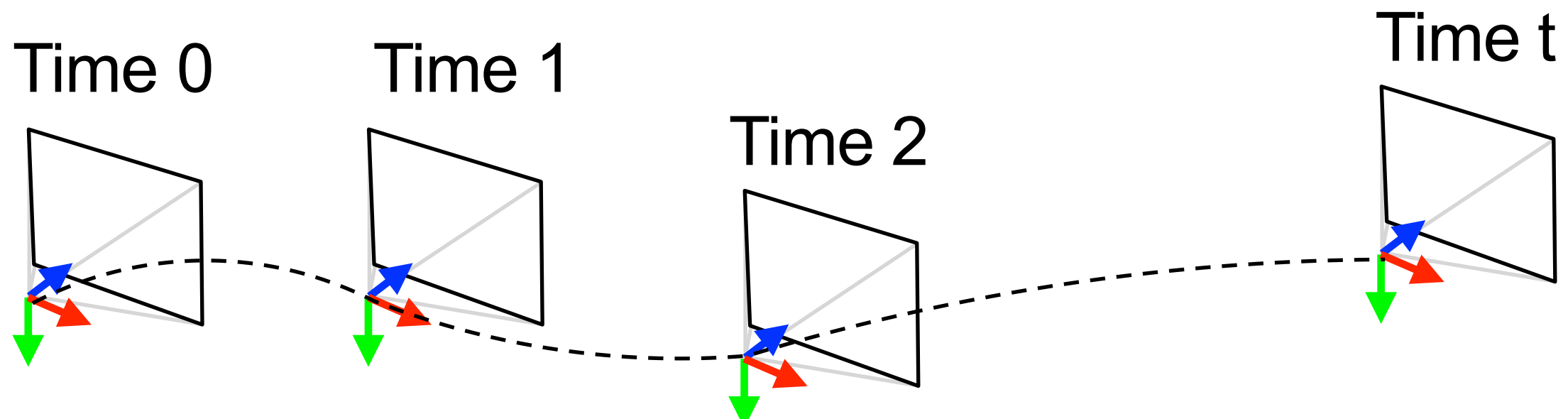
Meshes



Map representation	3D Topology ?	Lightweight?	Filters Noise/ Outliers?	Semantics?
Point Clouds	✗	✓/✗ No, if Dense	✗	✓/✗ No, if Sparse
primitives & objects	✗	✓	✓	✓/✗ No, if Sparse
Voxels	✓	✓/✗ No, if small voxel	✓	✓/✗ No, if large voxel
3D Mesh	✓	✓	✗	✓

Multi-view Stereo

From previous lectures: we know how to use SLAM to get a good estimate of the poses of the cameras



Stereo



Multi-view stereo

Multi-view Stereo

Towards Internet-scale Multi-view Stereo

CVPR 2010

Yasutaka Furukawa¹ Brian Curless²

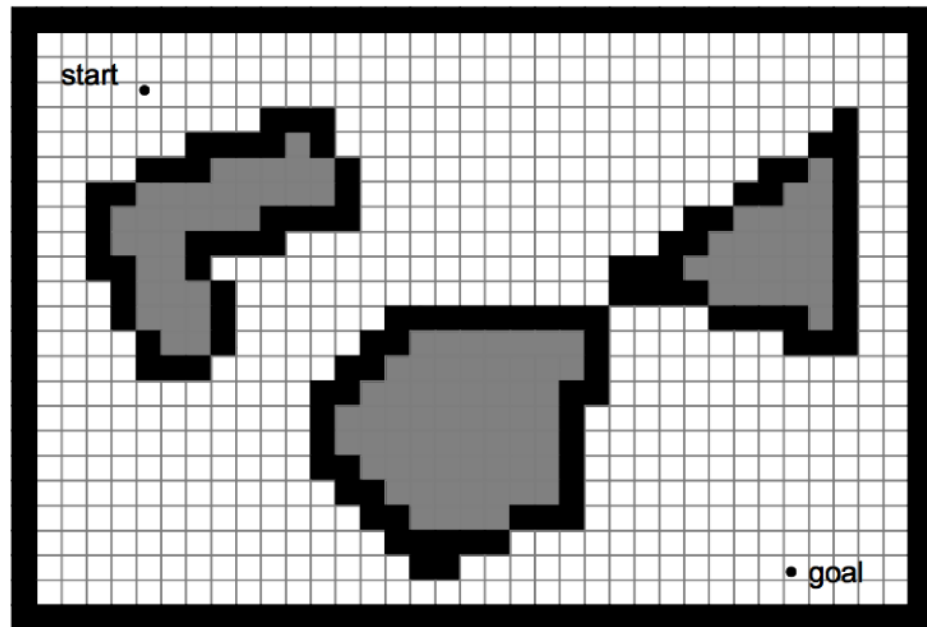
Steven M. Seitz^{1,2} Richard Szeliski³

Google Inc.¹

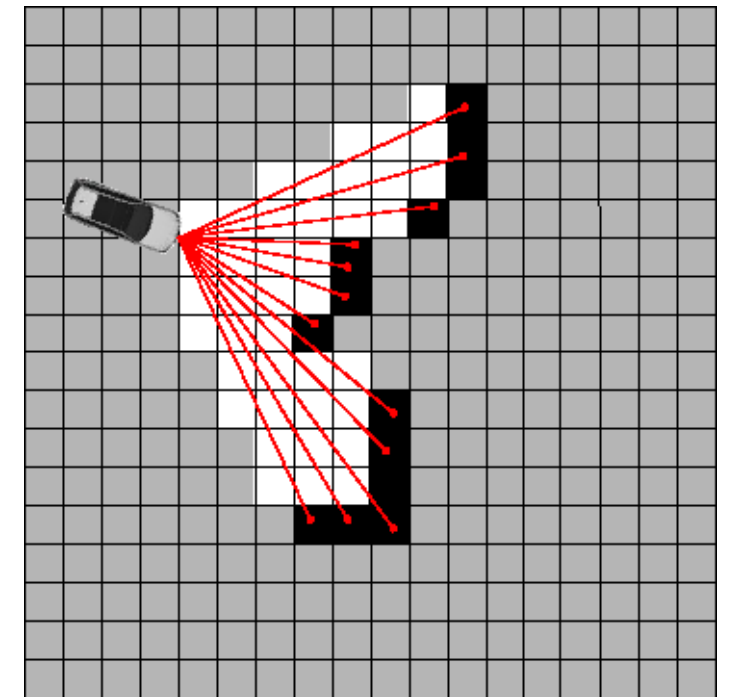
University of Washington²

Microsoft Research³

A Gentle Start: 2D Occupancy Grid Maps



□ = free ■ = obstacle ■ = unknown



- discretize the environment into cells
- Each cell holds real number $[0,1]$, representing the probability of the cell being occupied

Map posterior

$$p(m \mid z_{1:t}, x_{1:t})$$

Unknown
Map

Known sensor
Depth and robot poses

A Gentle Start: 2D Occupancy Grid Maps

$$p(m \mid z_{1:t}, x_{1:t}) \xrightarrow{\text{blue arrow}} p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})$$

↓
 Binary value
 (free/occupied)

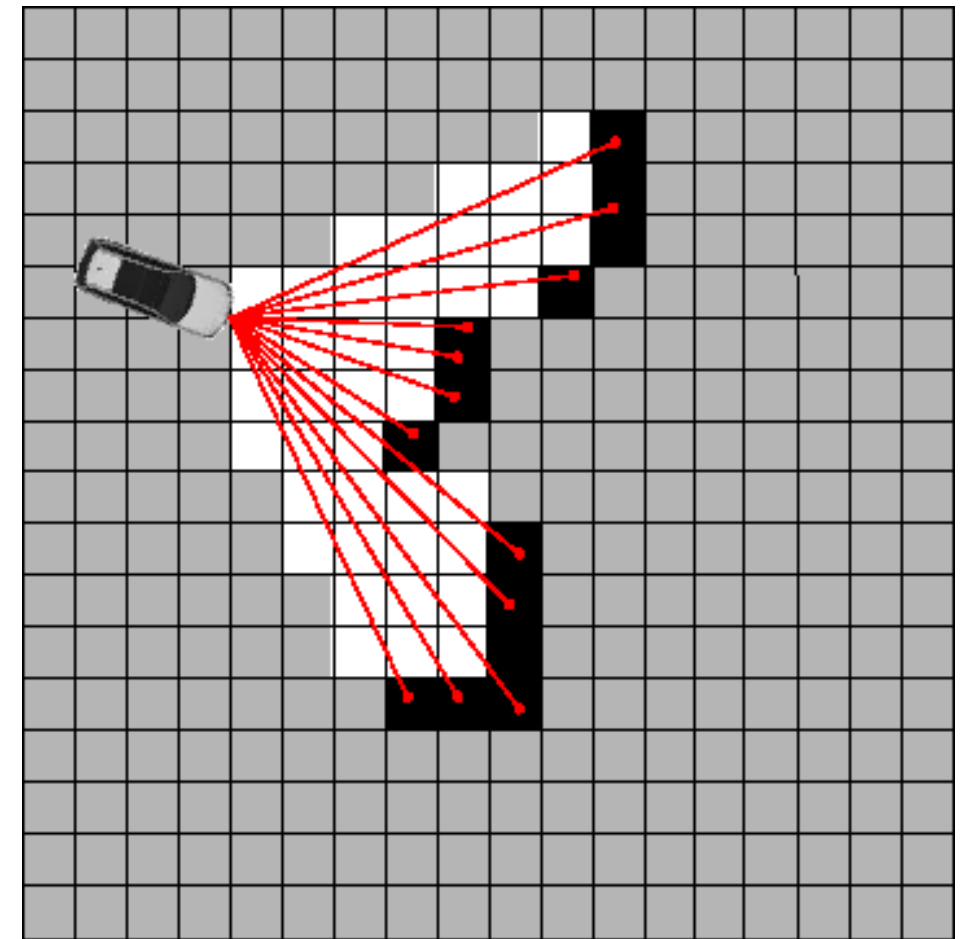
Probability
 of cell
 being occupied

Bayes rule (omitting “x” for simplicity):

$$p(m_i \mid z_{1:t+1}) = \frac{p(z_{t+1} \mid m_i) p(m_i \mid z_{1:t})}{p(m_i)}$$

↘
 Uninformative
 Prior

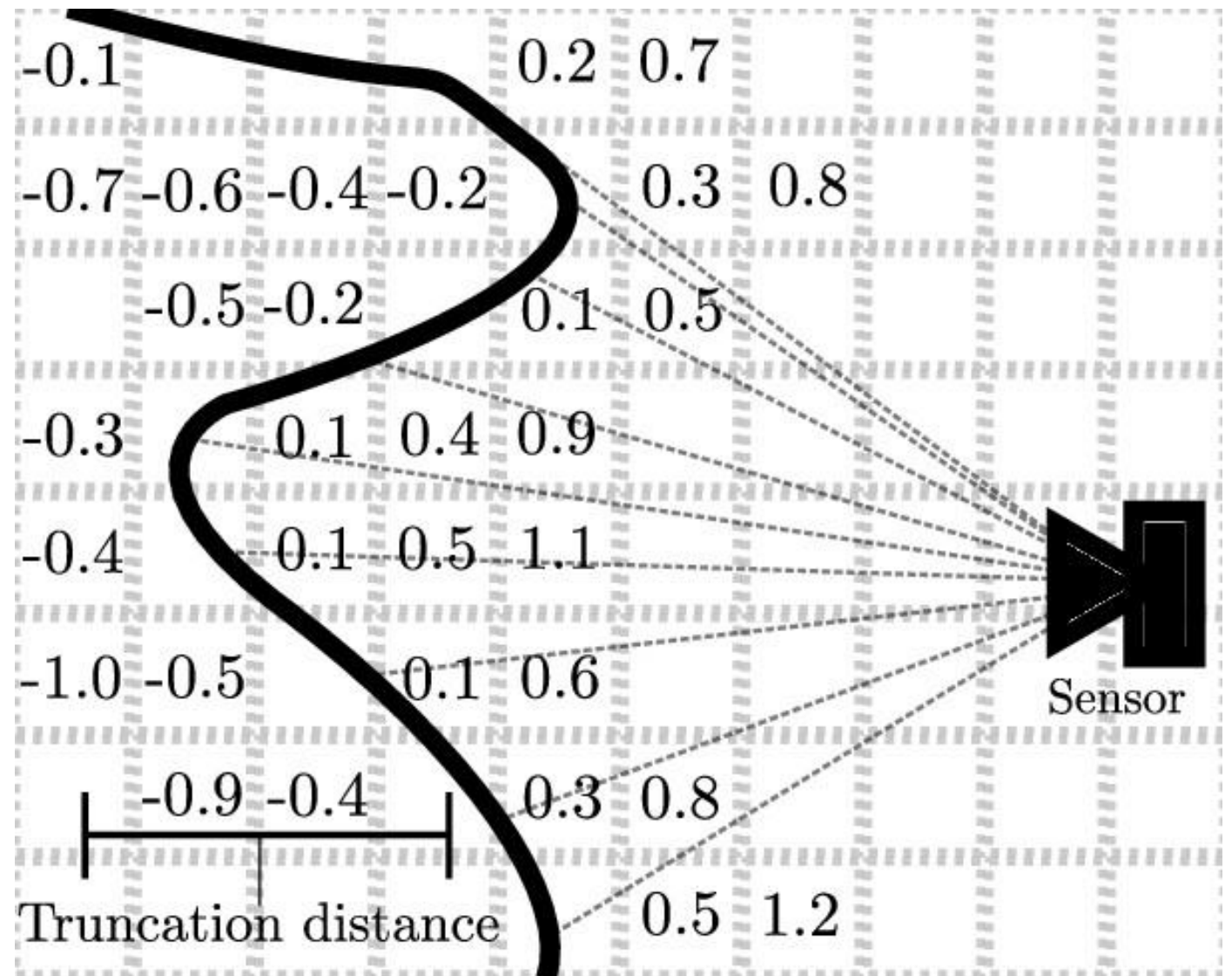
Log-odd representation is typically used to avoid numerical instabilities



$$\frac{p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})}{1 - p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})} \xrightarrow{\text{blue arrow}} l_{t,i} = \log \frac{p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})}{1 - p(\mathbf{m}_i \mid z_{1:t}, x_{1:t})}$$

Truncated Signed Distance Function (SDF)

- Store distance to nearest obstacle (with sign)
 - Only update around obstacle itself
- (implicit surface model)



Update rule*:

$$d(\mathbf{x}, \mathbf{p}, \mathbf{s}) = \|\mathbf{p} - \mathbf{x}\| \text{sign}((\mathbf{p} - \mathbf{x}) \bullet (\mathbf{p} - \mathbf{s})) \quad (1)$$

$$w_{\text{const}}(\mathbf{x}, \mathbf{p}) = 1 \quad (2)$$

$$D_{i+1}(\mathbf{x}, \mathbf{p}) = \frac{W_i(\mathbf{x})D_i(\mathbf{x}) + w(\mathbf{x}, \mathbf{p})d(\mathbf{x}, \mathbf{p})}{W_i(\mathbf{x}) + w(\mathbf{x}, \mathbf{p})} \quad (3)$$

$$W_{i+1}(\mathbf{x}, \mathbf{p}) = \min(W_i(\mathbf{x}) + w(\mathbf{x}, \mathbf{p}), W_{\text{max}}) \quad (4)$$

[Curless and Levoy, "A Volumetric Method for Building Complex Models from Range Images", 2007]

Kinect Fusion (2011)



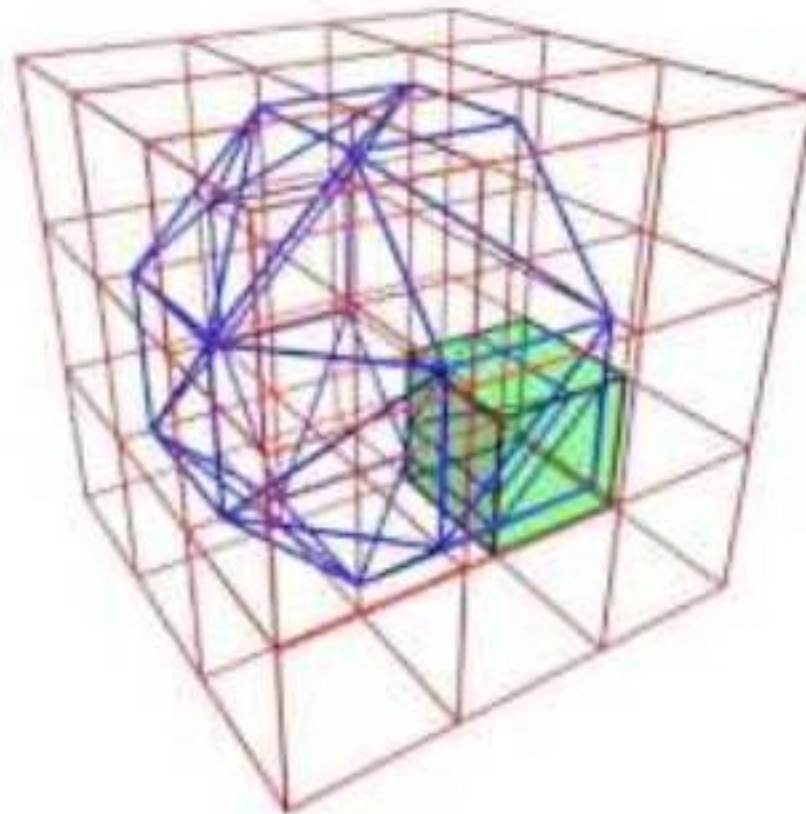
**3D reconstruction
(surface normals)**



**3d reconstruction
(texture mapped)**

From Voxels to Meshes

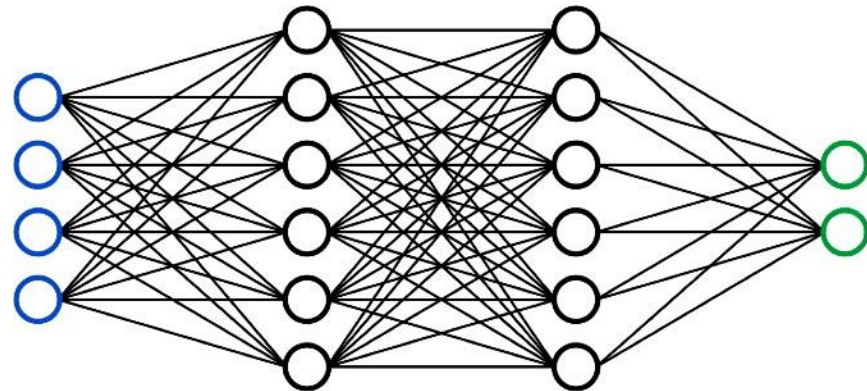
Implementation



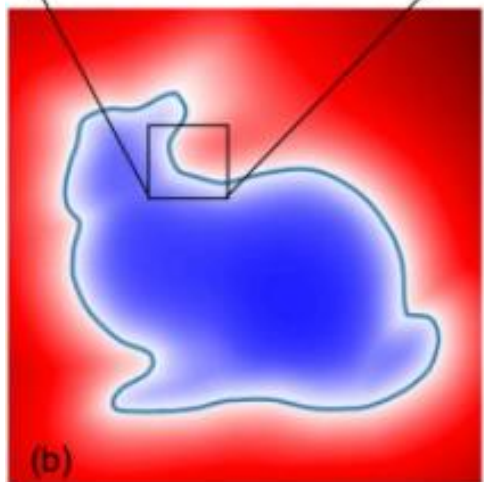
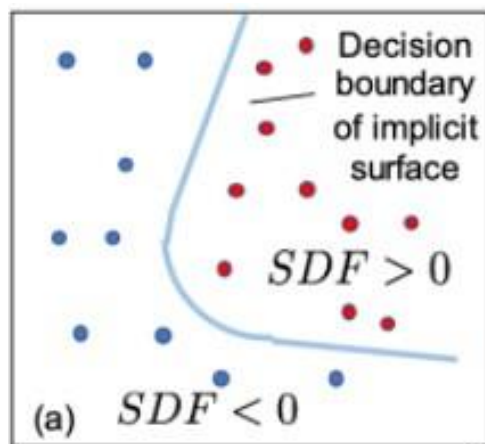
New Representations: Neural Implicit Surfaces

Use neural networks to define an **implicit** surface representation

(x, y, z)



Signed distance
To obstacles



(c)



DeepSDF: Learning Continuous Signed Distance Functions for Shape Representation, Jeong Joon Park, Peter Florence, Julian Straub, Richard Newcombe, and Steven Lovegrove, CVPR 2019.

Neural Radiance Fields or NeRF

Use neural networks to regress not a signed distance function, but **color -> rendering**

NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis

Ben Mildenhall*
UC Berkeley

Pratul P. Srinivasan*
UC Berkeley

Matthew Tancik*
UC Berkeley

Jonathan T. Barron
Google Research

Ravi Ramamoorthi
UC San Diego

Ren Ng
UC Berkeley

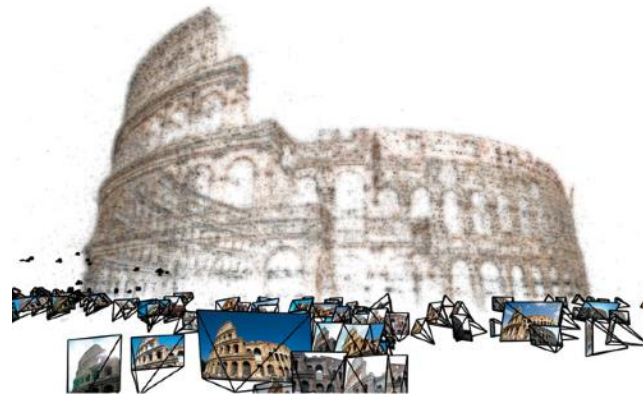
* Denotes Equal Contribution



Robot Perception or Computer Vision?

Computer vision

.. “a day on a cluster with 500 compute cores”



Robotics



50-100ms latency,
embedded,
incremental

No longer a dichotomy for many vision applications!

Robot Perception or Computer Vision?



Unordered
Vs
Sequential



Autonomous Navigation: Full stack

